Maxim Myshchak

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EDUCATION

Harvard University **Cambridge, MA**

B.S. Electrical Engineering / Computer Science **May 2030**

- Relevant Coursework: ES50 - Introduction to Electrical Engineering (circuits, hardware fundamentals), CS 50 - Introduction to Computer Science (C, Python, SQL), Math 21a - Multivariable Calculus, Physics 15a - Introductory to Mechanics and Special Relativity

PROFESSIONAL EXPERIENCE

NYC Department of Youth and Community Development - *Software Development Intern* **July 2026 - August 2026**

- Led development of a centralized, Python-driven test-case tracking and data visualization system for an 11-person Agile intern team conducting WCAG 2.2 accessibility audits and validation across 13 DYCD production web applications (23,000+ test cases)
- Wrote Python scripts (pandas, NumPy, matplotlib) to automate data validation, aggregation, and dynamic chart generation, giving management real-time visibility into testing status, increasing actionable insights by 70%
- Debugged WCAG 2.2 accessibility defects in discoverDYCD webpage, leveraging Angular, pushing code to production on Azure
- Authored technical architecture documentation (Mermaid diagrams) mapping application workflows and service integrations for cross-functional engineering teams

GE Aerospace Explore Engineering Program - *Intern* **November 2025 - February 2026**

- Analyzed and compared diverse energy sources for propulsion systems, including cost, energy density, power, and storage requirements
- Conducted in-depth research on bypass ratio and compression ratio and their impact on turbofan engine design, including fan diameter, noise, installation considerations, material limitations, and emissions implications
- Developed a technical presentation to effectively communicate the limiting factors associated with bypass and compression ratios that helped guide the decision-making for a next-generation propulsion system at GE Aerospace

MIT Lincoln Laboratory - *Student Engineer / Researcher* **July 2025 - July 2025**

- Designed and executed Doppler RADAR experiments, applying signal processing and circuit fundamentals, presenting findings to laboratory researchers, scientists, and engineers
- Collaborated with 25 ambitious individuals from across the nation (3% national acceptance rate for this residential program) to debug hardware and overcome obstacles related to learning how RADAR works and challenges encountered when conducting experiments
- Acted as a visionary, unifier, and team member throughout the process, providing team with a background in physics and mathematics while mutually reinforcing communication skills and collaboration-centric research practices

New York City Makerspace - *Lead Designer & Fabricator* **July 2024 - August 2024**

- Designed and prototyped in CAD a CNC-milled adjustable standing desk for NYC students with ADHD; iterated based on user feedback and tested in NYC public school environments; being published to the Makers Making Change open-source repository
- Assisted in alleviating usable scrap wood from ending up in the landfill by repurposing lumber to construct 4 wooden chairs, later donating them to elders in the community in need of furniture

PROJECTS

HackHarvard - AI-Powered Crop Monitoring Drone for Small-Farm Sustainability **October 2026**

- Built a low-cost AI drone system enabling small farms that cannot efficiently monitor their crops to track crop health sustainably
- Trained a YOLOv8 deep learning object detection model on the Global Wheat Head dataset using GPU-accelerated training, then exported it to ONNX for edge AI inference, running OpenCV computer vision on the drone's live video feed to detect wheat heads
- Engineered embedded hardware on an Arduino UNO Q (C++, Python, Linux) with sensors, networking a drone, phone, and PC
- Integrated agentic AI and sensor fusion of crop scores with NOAA and NASA data to deliver actionable recommendations to farmers

LEADERSHIP EXPERIENCE

Science Olympiad Team - *Captain & Problem-Solver* **September 2023 - June 2026**

- Organized 30+ members into their respective events and ensured readiness for competitions, providing crucial engineering mentorship
- Managed the team's financial responsibilities, organizing the raising of over $1,600+ funds and Qualified for NY State Championships
- Achieved: **4th place** in Columbia University's **Integration Bee**, **4th** in Material Science (out of 75+ national teams); 6th place in Wind Power, 6th in Air Trajectory, 9th in Circuit Lab, 11th place in Optics, 13th in Robot Tour, 14th in Engineering CAD (out of 55+ teams)

Engineering/Computer Science Club - *President & Community Liaison* **September 2022 - June 2026**

- Revamped outreach, transforming club from a dormant group of 4 students into a 19-member hub for beginner-friendly workshops on 3D-printing, coding, and electronics - allowing students to develop assistive tech, ultimately earning National Finalist recognition
- Mobilized 4 teams for the MIT Cre[at]e Challenge, engineered hardware prototypes to address challenges for those with disabilities
- Collaborated alongside people with disabilities, taking in feedback to create improved iterations, later open-sourcing the prototypes

SKILLS/AWARDS

- **Coding Languages**: C, C++/Arduino, MATLAB, Python (Pandas, Matplotlib, NumPy), Java, JavaScript, HTML, CSS, C#, SQL
- **Computer Skills:** Linux OS, Embedded Linux, Git, Artificial Intelligence (AI), Machine Learning, Deep Learning, Computer Vision (OpenCV, ONNX, YOLOv8), Robotics, UDP/WiFi Networking, Azure Cloud, .NET, Angular, Figma, Microsoft Office, Google Suite
- **Engineering Skills:** Embedded Systems, Microcontrollers/Arduinos, I2C Sensor Integration, Sensor Data Acquisition, Soldering, Breadboarding, Lab Equipment (Oscilloscope, Multimeter, Logic Analyzer), Circuitry, Integrated Circuitry, Altium (PCB design), Power Systems, CAD (TinkerCAD, OnShape, Fusion360), Hardware Testing, Debugging & Validation
- **Languages:** English, Ukrainian, Spanish, and Russian